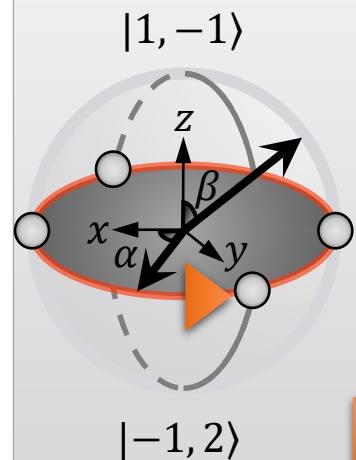
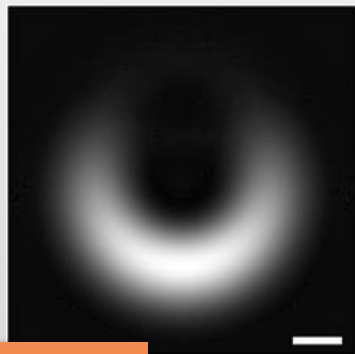


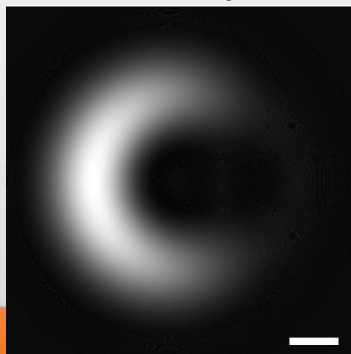
Gray exciton transition rate $\Gamma_{G,Q}^{1,-1,-1,2}(\mathbf{Q}; \alpha, \beta)$



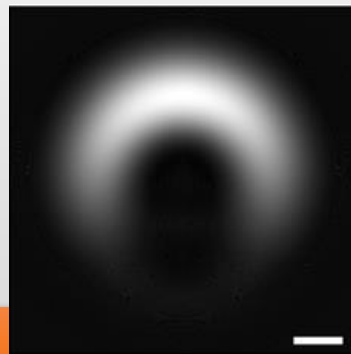
$\alpha = 0$



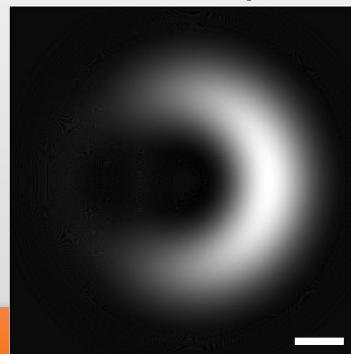
$\alpha = \pi/2$



$\alpha = \pi$



$\alpha = 3\pi/2$



Equator